

The fully emergent proton: drive with the residual, observe everything else an epic design note, for discussion

Tessera — cobordism subsystem
(epic proposal; builds on #457/#475, #489/#491/#549, #541)

Abstract

We propose an epic that finishes what the emergent-merge engine started: build the proton by extremizing the one objective we already have, $F = \|\nabla S_{\text{Regge}}\|^2 + \Gamma r_U$, where r_U pins *only the prepared initial state* (the experiment's boundary data), and treat **every other physical quantity — color content, flavor, spin, electric charge, mass, radius, even the intermediate event structure — as an observable read off the resulting lattice** after convergence. The current source of truth, `Proton::build()`, is most of the way there: topology grows from a single Δ^4 seed, geometry emerges from the relaxation, nothing is welded or hand-placed. What remains imposed is a short list — the two-step event graph with its colored intermediate targets, the singlet output target, the answer-shaped convergence gate, and the all-spacelike seed — and the record says each item can be removed outright or demoted to a read. The strongest argument is empirical, not aesthetic: hand-imposed symmetry *provably deleted* the flavor observable (#414), and the emergent register *delivered* it (#475). The design is four independently falsifiable rungs — the joint node, the inputs-only objective, Lorentzian activation, and the entangled readout battery — each with pre-registered pass/fail criteria whose negatives are results.

1 The goal and the objective

The objective, stated once (the full *complex* Sorkin/dual Regge action — keep Im; extremize $\delta S = 0$, never minimize $|S|$):

$$F = \|\nabla S_{\text{Regge}}\|^2 + \Gamma \cdot r_U, \quad \|\nabla S_{\text{Regge}}\|^2 = \sum_e \left| \frac{\partial S}{\partial \ell_e^2} \right|^2, \quad (1)$$

with r_U the relabeling-invariant, zero-filled `residualForPeriods` over the emergent register holes. The epic's thesis: Eq. (1) with r_U restricted to the *prepared inputs* is the entire drive; the proton is whatever stable structure emerges, and everything we want to know about it is a measurement on the relaxed lattice.

This note builds directly on the `MultiCobordism` engine (#491/#549), the T5 emergent-optimizer architecture and findings (#462/#475), the emergent-intermediates bracket (#434/#438), the spin/readout theory (`docs/theory/cobordism/proton-spin/`), and the Lorentzian-dormancy investigation (#541).

2 The constitution: what is allowed to shape the lattice

Unchanged from the T5 design note, restated as this epic's ground rules. The *only* conditions imposed on any build are:

1. the **manifold gate** — every move passes `dualComplexValid` (surgery is allowed *because* it is gated);
2. **extremization of F** — greedy ΔF moves plus the trap door, then the Stage-2 Wirtinger relaxation; the action stays complex end to end;
3. the **prepared initial state**, held representable through its r_U term (never by freezing vertices — `MultiCobordism::pinnedBoundaryVertices` is deliberately empty, `MultiCobordism.cpp:292`);
4. **room to act** — the single- Δ^4 seed plus the gated **precone** growth knob.

Everything else — a target topology, a b_k goal, a move recipe, a frame condition, and (the subject of this epic) a *final-state target* — is either forbidden or scheduled for demotion to a post-hoc observable. Every observable claimed on a converged structure must pass the GAUGE and RELABEL invariance gates (the #485 convention).

3 What `Proton.cpp` still imposes, precisely

#	imposition	where	tension with the goal
1	The two-step event graph with colored intermediate targets: step A drives 2 neutral pairs $\rightarrow$ diquark $\{1, \omega\} \sqcup$ antidiquark $\{1, \omega^2\}$	<code>Proton.cpp:67-92, : 143-150</code>	the event structure and a colored intermediate are placed in the objective, not observed
2	The hand-off between the steps is nominal : step B starts a <i>fresh</i> Δ^4 seed and imposes the <i>ideal</i> diquark + third quark $\{\omega^2\}$ as its inputs; step A's emergent output is never read out and fed forward (only its r_U is reported)	<code>Proton.cpp:94-114, : 150</code>	as a physical event chain this is not composition (the result-state = next-boundary rule is not exercised)
3	The proton singlet $[1, \omega, \omega^2]$ as the output target , read on the whole cobordism	<code>Proton.cpp:107-109; whole-read branch <code>MultiCobordism.cpp:266-273</code></code>	the answer sits in the objective — held by r_U (the sanctioned knob), but a drive nonetheless
4	The convergence gate <code>colorResidual < colorTolerance $\wedge$ holes $\geq$ minQuarkHoles</code> (defaults 0.5 / 3)	<code>Proton.cpp:159-160, Proton.h:86-89</code>	selection by the known answer instead of by stationarity + persistence
5	The all-spacelike seed ($\ell^2 = +1$ everywhere)	<code>Proton.cpp:62-63</code>	not a target, but it silently pins the dynamics Euclidean (§4.5) — half the observable battery then reads zero for structural reasons

Impositions that are **legitimate and stay**: the prepared inputs (neutral $\Sigma = 0$ $q\bar{q}$ pairs — the choice of experiment), the objective itself, the gate, and the seed-size knobs.

4 The evidence for de-imposition

4.1 De-imposition is what unlocked flavor. On the hand-symmetric A_4 register, flavor is *provably* unreadable: any relabeling-invariant per-window measurable commutes with the transitive window-cycling g and is constant on the orbit — measured discriminator margin $\sim 3 \times 10^{-6}$, machine zero (`proton_flavor_split.md`, #414). On the **emergent, non-symmetric** register the same read succeeds: per-hole Dirac–Kähler charge distinguishable with relative spread 0.52–0.61, persisting (often sharpening) through Stage-2 relaxation in 20/21 structures (`t5_emergent_optimizer_findings.md`, #475). The lesson generalizes: hand-imposed symmetry deletes observables; emergence carries them.

4.2 The dynamics does not choose the staged diquark path. With both endpoints pinned and the interior free, the emergent intermediate is nearly neutral (singlet overlap 0.995, colored $\sigma \approx 0.096$); the imposed strong- $\bar{\mathbf{3}}$ diquark is *hosted* (connected-bulk carry $r_U = 0.137$, $\|\nabla S_{\text{Regge}}\|^2 = 49.1 < 100$) but *diverges* from the emergent path — overlap 0.519 against the pre-registered 0.90 (`438-fixed-bipartite-sequence-f5ad527.md`).

Caveat. That bracket ran on the retired 2+1 D base; the numbers may not transfer. The emergent-vs-imposed methodology does.

4.3 The diquark is recoverable as a read, so it need not be a target. The pair-loops are Poincaré duals of the complementary holes — $[\gamma_{ij}] = -[k]$ — so the diquark *is* the dual of the spectator-quark hole, and the $u:u:d = 2:1$ multiplicity is a dual-basis statement on the finished register (`pair_loop_quarks.tex`, with a concrete falsifiable test: the odd-one-out loop must coincide with the diquark loop, under the GAUGE/RELABEL gates). Removing the staged intermediate from the drive loses nothing that cannot be read afterwards.

4.4 The engine already supports every shape this epic needs. N input blocks (the ctor takes lists); the #489 joint build (3 neutral pairs $\rightarrow$ [proton, antiproton] as two localized output blocks) ran on this engine and delivered the carried singlet *and* independent per-hole flavor; and with **outputTargets empty the objective is inputs-only today** — the `rU` else-branch adds no output term when no output blocks exist (`MultiCobordism.cpp:274–284`). Rung 2 (§5) is mechanically a constructor argument, not an engine change.

4.5 The Lorentzian sector is dormant by dynamics, not by physics. The converged proton has $\text{Im } \ell^2 \equiv 0$, $\min \text{Re } \ell^2 = 0.262$ — the Stage-2 clamp ($[0.05, 20]$, `MultiCobordism.cpp:672–676`) never fires; the all-spacelike seed simply keeps the descent in the Euclidean basin (`541-why-geometry-spacelike-30047e0.md`). Meanwhile every electric observable is measured to be Lorentzian-gated: emergent Gauss-law charge $|Q_e| = 0.297 / 0.117$ with timelike worldlines vs **exactly** 0 on the all-spacelike control (#434/#438), and the closed-surface flux is topologically protected ($\sim 10^{-16}$) either way. Charge, E/B , photons-as-null-edges, temporal curvature, and the transition-amplitude program all wait on this.

4.6 The remaining spin gap is a readout, not a build defect. The build creates the entanglement; the per-hole extraction discards it. The product of per-hole spinors *provably* floors at the uncorrelated mixture ($J^2 \sim 9/4$; the entangled proton value $3/4$ is unreachable by any Bloch-vector reduction), while the instrument itself is validated exact on clean states ($3/4 / 7/4 / 15/4$) —

`spin_readout.tex`, `joint_proton_spin_findings.md` (#489). The fix is the total-space operator $J_a = \sum_h P_h \Sigma_a P_h$ whose **cross-hole terms** carry the entanglement (#477), on the fiber $\leftrightarrow$ cells lift (#495); `cartan_weyl_gluon.tex` supplies the gauge-invariant chamber coordinate (the connected correlator C_{ij}) and four pre-scoped experiments, including the decisive “ C_{ij} two ways” test and the holonomy-sourced chamber point (the operational gluon test).

4.7 What “carried vs floored” already gives us for free. Neutral pairs carry ($r_U \sim 10^{-8}$, $\|\nabla S_{\text{Regge}}\|^2$ at its ~ 88 floor) where free/colored quarks floor ($r_U \sim 0.5$, $\|\nabla S_{\text{Regge}}\|^2 \sim 14600$) — confinement as a property of the objective, established in the merge track. Observation mode leans on exactly this asymmetry: the drive never says “singlet”, but the landscape does.

5 The design: four rungs, each independently falsifiable

5.1 Rung 1 — collapse the two-step into one joint node (the #489 shape)

One MultiCobordism: inputs = **three neutral $q\bar{q}$ pairs**, outputs = **two localized blocks** (baryon $\sqcup$ conjugate antibaryon — the multi-output branch of `rU`, `MultiCobordism.cpp:274-278`), on a single Δ^4 seed, driven by the same Grow $\rightarrow$ directed probes $\rightarrow$ Evolve $\rightarrow$ Relax sequence `build()` uses today. The two-step build is **demoted to the reference oracle** (the #454 pattern), kept for A-vs-B comparison. The intermediates become reads: the pair-loop dual-basis flavor test (§4.3) and the mid-build register content replace the imposed diquark target.

Why two output blocks and not singlet-on-the-whole: with three pairs the whole complex must also host the conjugate sector; a single whole-read target $[1, \omega, \omega^2]$ would fight the antibaryon’s $[1, \bar{\omega}, \bar{\omega}^2]$ periods. The $2 \rightarrow 2$ recombination branch already models exactly this.

Discussion item: if the pairwise-microcausality argument in `Proton.h` is to be preserved as a physical claim, the alternative is to make the two-step’s composition real — feed step A’s **measured** emergent output forward as step B’s boundary instead of the ideal constants. Both arms can run; the observables adjudicate.

5.2 Rung 2 — retire the output targets; convergence = stationarity + persistence

Same node, **outputTargets** = $\{\}$ — the objective is $\|\nabla S_{\text{Regge}}\|^2 + \Gamma \sum_i w r_U(\text{input}_i)$, nothing else. The convergence gate (§3, item 4) is replaced by:

- **stationarity** — Stage 2 stops on its relative-tolerance test (`lastStage2Stationary()`), and no Stage-1 candidate move lowers F ;
- **persistence** — the claimed structure survives continued Evolve/Relax passes and fresh-seed re-relaxations with its observables stable (the Stage-2 stability-sweep methodology from #475, promoted to the acceptance criterion).

Then **observe** (§6): hole count, singlet content *as a diagnostic* (`residualOfTargetStateAgainstHarmonic` already exists as a static — used after the fact, it is a measurement, not a drive), σ_R , per-hole DK charge spread, spin, charge, mass, radius.

The claim under test, pre-registered: *the stable endpoint of the inputs-only drive is a bound, localized, color-neutral ≥ 3 -hole object with a conjugate partner — a baryon pair — rather than three identity-transported pairs*. This is genuinely open, and the record names the failure modes honestly: single-ended pinning under-constrained the isolated creation node (conformal drift, color spread *worsening* $0.47 \rightarrow 2.44$, #435), and merge transport tends to identity (#376). Both outcomes are results (§7).

5.3 Rung L (parallel) — activate the Lorentzian sector

Per #541’s recommendation: (a) replace the hard $\text{Re } \ell^2 \geq 0.05$ floor with a **causal-aware degeneracy guard** — forbid only the light-cone band $|\ell^2| < \varepsilon$, both signs, so timelike edges are admissible and degeneracy still is not; (b) put the time direction in the **boundary data** — timelike worldline edges on the prepared pair inputs (a cobordism has a time direction; preparation, not answer-insertion). Verify the complex deficit / Lorentzian volume path across signature changes, and watch the conformal runaway by its restoring force (the standing rule: never sidestep it). Null edges that appear are the photon candidates, and they should be *sourced*, not spontaneous.

Unlocks, with their already-built readers: $Q = \oint_S E$ (**gaussLawCharge** — the genuine holonomy, protected to $\sim 10^{-16}$, vs the drifting hand-weighted covector), **fieldStrengthSplit** (E/B), the CPT pairing $Q_p + Q_{\bar{p}} = 0$ across the emergent conjugate sectors, the temporal curvature panel, and the action-coupled amplitude Experiment B of **transition_amplitudes_from_relaxation.tex** (does $\delta S = 0$ protect unitarity under backreaction — either answer is a result).

5.4 Rung R (parallel) — finish the entangled readout battery

Pure reads, no new drive terms, each gated GAUGE/RELABEL:

1. **#495 fiber↔cells lift** (Whitney/Kähler–Atiyah) — the last rung between the validated instrument and a J^2 read off the emergent proton;
2. **#477 total-space J^2** with the cross-hole terms (the entanglement carriers);
3. C_{ij} **chamber coordinate** + the two cheap **cartan_weyl_gluon** experiments (instrument reproduction; C_{ij} -two-ways — vertical vs holonomy routes, decisive either way);
4. **holonomy-sourced chamber point** — the transport intertwiner M as the entangling source; the relaxed proton lands on the proton corner without the chamber point being put in by hand (the operational meaning of “the exponential acts like a gluon”);
5. **pair-loop dual-basis flavor** on the built proton (2:1 multiplicity; odd-one-out = diquark loop);
6. **mass/radius on the relaxed interior**, the #451 methodology ported to 4D: closed-fan hinges only, skeleton built in C++, dual-volume radius + shell deficit mass, $r \cdot m$ reported with its definitional spread (order-of-magnitude claim only).

5.5 Search practicality (all rungs)

Removing targets flattens the ΔF landscape — T5’s cautionary tale: the first cone-out often raises F , and an undirected walk never left $b_2 = 0$ in $\sim 13.9\text{M}$ gated moves. The levers are in-tree: the trap door, restarts, the incremental ΔF , the directed probes, and the RL harness. Two consistency notes. **directedConeOut** scores this node’s **rU**, which in rung 2 contains only the input terms — expect weaker guidance; that is honest, and it is what the policy is for. And the RL reward for the emergent arm must be the engine’s F **alone** — the current **rl_objective_search.py** shaping (**hole_reward_weight**, **rstate_reward_weight**) is fine for the oracle arm but would smuggle the proton into the policy on the emergent arm. RL work stays on the examples-not-tests lane.

6 The observable battery (read after convergence; never loop conditions)

quantity	reader	status
color content / confinement	<code>residualOfTargetStateAgainstHarmonic</code> as diagnostic; σ_R ; carried-vs-floored contrast	ready
flavor	per-hole DK charge spread on the emergent register; pair-loop dual-basis test	ready / rung R.5
spin (baryon content)	per-hole $ \langle S \rangle = \frac{1}{2}$, product J^2 floor 3/2 (dual-edge frame)	ready
spin ($\frac{1}{2}$ vs $\frac{3}{2}$)	total-space J^2 with cross-hole terms; C_{ij} chamber coordinate	rung R.1–R.3
electric charge	$Q = \oint_S E$ Gauss-law holonomy; CPT pairing across conjugate sectors	blocked on rung L
photon	null-edge census (sourced, not spontaneous)	blocked on rung L
mass / radius	interior shell deficit; dual-volume radius; $r \cdot m$ (order-of-magnitude)	rung R.6
event structure / intermediates	mid-build register content; pair-loop duality	rung 1
transition amplitude / unitarity	$Z_T(q, \alpha) = (T^{-1}q)^\dagger G \alpha$, $u' = (T^\dagger)^{-1}$, $\{G = I, T^\dagger T = I\}$ certificates	Exp. A ready; B on rung L

7 Pre-registered criteria, and what each negative would mean

Thresholds set before running, never loosened; negatives are reported as findings (the #434/#438 discipline).

Rung 1. The joint node converges at least as often as the two-step across a fixed seed battery; carried singlet on the baryon block; independent per-hole flavor (spread ≥ 0.1); pair-loop test passes. *Negative* $\rightarrow$ the two-step’s staging is load-bearing; keep it and make its composition real (§5.1, discussion item).

Rung 2. Stationary + persistent endpoint with ≥ 3 emergent holes on the baryon cluster, singlet diagnostic $\leq$ the rung-1 carried level, conjugate pairing observed. *Negative (identity transport / no binding)* $\rightarrow$ the final-state pin is demonstrably the minimal necessary imposition — quantified, and the singlet-as-knob (today’s `Proton.cpp`) stands as the documented fallback. *Negative (conformal drift)* $\rightarrow$ the restoring-force diagnosis from #435/#434 applies; report which term failed to regulate, do not bolt on a boundary.

Rung L. Timelike content survives relaxation (no clamp artifacts — the guard, not the floor, binds); $|Q_e| > 0.05$ with the all-spacelike control at 0; flux protection and CPT pairing hold. *Negative* $\rightarrow$ the Euclidean basin is deeper than the seeding; report the basin boundary, revisit boundary data (not the objective).

Rung R. Instrument checks reproduce 3/4 / 7/4 / 15/4 exactly; then the emergent read either moves J^2 decisively below 3/2 toward 3/4 (proton channel) or lands at 15/4 (Δ) — *either definite channel is a success*; an indefinite $\sim 9/4$ after the cross-hole terms would falsify the total-space-operator hypothesis itself, which is worth knowing.

Control (suggested). Run rung 2 with **two** pairs — does a meson-shaped 2-hole neutral object emerge? Distinguishes “binding is generic under F ” from “three is special” (the bipartite

obstruction says the singlet is irreducibly 3-valent; a meson control probes the complementary claim).

8 Out of scope

- The codim-2 surgical move / b_2 register (proven unreachable and mooted — flavor reads at $k = 3$).
- Any auxiliary term or rule in F (b_k goals, harmonic-dimension policing, frame conditions).
- Retiring the hole-based period readout cores or other load-bearing API refactors.
- Docs/animation work beyond the findings notes each rung owes.

9 Proposed epic decomposition (tickets cut after sign-off)

ticket	rung	deliverable
E1	1	joint formation node as a Proton build mode driving the #489 shape; two-step demoted to reference oracle; A-vs-B harness + findings note
E2	1	pair-loop dual-basis flavor read on the built proton (GAUGE/RELABEL-gated)
E3	2	inputs-only build (outputTargets = {}); stationarity + persistence acceptance; observable-battery report; meson control
E4	L	causal-aware degeneracy guard replacing the $\text{Re } \ell^2$ floor; Lorentzian boundary data on the pair worldlines; signature-change verification
E5	L	charge battery on the Lorentzian build: $Q = \oint E$, E/B , null-edge census, CPT pairing across conjugate sectors
E6	R	#495 fiber $\leftrightarrow$ cells lift + #477 total-space J^2 (C++ on MultiCobordism/EigenstateSynthesis)
E7	R	C_{ij} chamber readout: instrument check + C_{ij} -two-ways + emergent read
E8	R	holonomy-sourced chamber point (the gluon test)
E9	R	mass/radius battery on the relaxed 4D interior (#451 methodology)
E10	—	RL emergent arm: reward = engine F only; benchmark vs random/grow-only on E3 (examples lane)
E11	L	transition-amplitude Experiment A on the equivariant fill, then B on the relaxed build (unitarity under backreaction)

Dependencies: $E1 \rightarrow E2/E3$; $E4 \rightarrow E5/E11\text{-B}$; $E6 \rightarrow E7 \rightarrow E8$; E9, E10 ride on E1/E3. Suggested order of first bites: E1 + E4 in parallel (E1 is mostly configuration; E4 is the physics unlock), E2 immediately after E1 (cheap, pure read).

10 Open questions for the discussion

Resolved in discussion. Three decisions are settled and the first work item is cut as #555: (i) **Proton.cpp** stays **canonical and untouched** — the emergent arm is a side-by-side sibling, **ProtonIngredients**: the identical two-step drive with *only* step B’s singlet output target dropped (**outputTargets** empty), so exactly one variable differs from canonical — a minimal-delta A/B rather than the rung-1 joint node, which stays a future rung. Its animation (**examples/cobordism/emergent_proton.py**) reuses the canonical charts verbatim. (ii) The epic is cut only after

the rungs are discussed individually; #555 stands alone until then. (iii) Question 4 is answered in the negative for *initialization*: **no timelike edges are initialized**. At initialization no time has passed — causal structure marks sequences of events (in the causal-set sense) and may only *emerge*. Rung L therefore reduces to the causal-aware degeneracy guard plus whatever causal content the dynamics produces; Lorentzian *boundary data* is off the table for the emergent arm. Question 1’s composition-real variant and questions 2, 3, and 5 remain open.

1. **Two-step disposition** — oracle only, or also make its composition real (feed the measured step-A output forward)? Keeping both arms is cheap; the header’s physical claim (`Proton.h:22-33`) deserves an explicit verdict rather than silent retirement.
2. **Input specification for the joint node** — the $\mathbb{Z}_3$ -symmetric neutral triple ($\{1, -1, 0\}$, $\{0, 1, -1\}$, $\{-1, 0, 1\}$) vs the current pair set; and how much `inputResidualWeight` / Γ re-tuning rung 2 needs once the output terms vanish from `rU`’s scale.
3. **Persistence protocol** — how many continued passes / fresh-seed re-relaxations, and which observables must be stable at what tolerance, for “persistent” to be claimed (candidate: the #475 sweep protocol verbatim).
4. **Lorentzian boundary data** — timelike ℓ^2 on which edges exactly (the pair worldlines only, or the whole seed given a CDT-style time direction), and the guard band ε .
5. **Claim language** — at what point do we say “proton” rather than “baryon-shaped bound state”? Proposal: “proton” requires the rung-R channel read ($J^2 \rightarrow 3/4$) *and* the charge read (+1 vs the neutron’s 0), both of which are observations, not targets.

References

- `src/cobordism/Proton.cpp`, `include/cobordism/Proton.h` — the current two-step source of truth.
- `src/cobordism/MultiCobordism.cpp` (`rU:257`, `pinnedBoundaryVertices:292`, `Stage-2 clamp:672`), `include/cobordism/MultiCobordism.h` — the engine.
- `docs/design/t5_emergent_optimizer_design.md`, `t5_emergent_optimizer_findings.md` — the architecture and the emergent-flavor result.
- `docs/design/434-emergent-intermediates-f1fb86d.md`, `438-fixed-bipartite-sequence-f5ad527.md`, `435-bipartite-creation.md` — the emergent-vs-imposed bracket and the under-constraint lesson.
- `docs/design/541-why-geometry-spacelike-30047e0.md` — the Lorentzian dormancy finding and fix.
- `docs/design/proton_flavor_split.md`, `proton_charge_gauss_law.md`, `proton_electroweak_finding.md`, `proton_radius_finding.md`, `451-proton-geometry-0a1a5aa.md` — the observable readers and their obstructions.
- `docs/design/composite_proton_spin_findings.md`, `joint_proton_spin_findings.md`, `docs/theory/cobordism/proton-spin/spin_readout.tex`, `pair_loop_quarks.tex`, `cartan_weyl_gluon.tex` — the spin program.

- `docs/design/transition_amplitudes_from_relaxation.tex` — the amplitude/unitarity layer.
- `docs/design/proton_bipartite_obstruction.tex`, `proton_symmetric_windows.tex`, `proton_session_review.tex` — the representation-theoretic ground (ϵ_{ijk} is irreducibly 3-valent) and the symmetric-window era.
- `examples/cobordism/rl_objective_search.py` — the RL harness whose reward the emergent arm restricts to F .